PH.48111 **FALL 2026**

Autonomous Theory Construction Laboratory

Lecture 13: Holdout Evaluation

Garam Yun 2026-11-23–2026-11-29

Overview

A previously hidden observation is admitted only after the model and decision rule have been frozen.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$D_h \in Vault(t < t_*)$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$Registry_{v+1} = Registry_v \cup O_h$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$L_{v+1} = L_v - 2 \ln p(O_h|T)$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Holdout Admission Rules
- Information-Flow Review IFA-5
- Yun, ch. 13